

and the most cost-effective to manufacture. CdTe panels perform significantly better in high temperatures and low-light conditions.

Amorphous silicon. Amorphous silicon is the non-crystalline form of silicon and was the first thin-film material to yield a commercial product, first used in consumer items such as calculators. It can be deposited in thin layers onto a variety of surfaces and offers lower

costs than traditional crystalline silicon, though it is less efficient at converting sunlight into electricity.

CIGS. CIGS is a compound semiconductor that can be deposited onto many different materials. It has only recently become available for small commercial applications, and is considered a developing PV technology.

says Semiconductor Equipment and Materials International (SEMI) president Debashish Paul Choudhury.

Under NTPC's Vidyut Vyapar Nigam Ltd (NVVN) scheme, there has to be a minimum capital investment of about Rs 750 million, at least 30 per cent of which has to come from the promoters' equity contribution. Thus most small entrepreneurs will not be able to take part in this scheme. However, outside of NVVN scheme, it is possible for the small entrepreneurs to get incentivised for putting up under-5MW solar PV power plants through generation-based incentives provided by Ministry of New and Renewable Energy.

How to go about?

"Upfront requirement for investment into a solar grid plant is possession of land suited for solar application, clear power purchase agreement with the local or state electrical authority and required finance," informs K. Subramanya, chief executive officer, Tata BPSolar.

The key building blocks of a grid-connected solar power plant include solar module (the power generating device), solar inverters (which convert the power produced into AC power) and power evacuation (which steps

up the voltage to the specified kilovolt level and injects into the grid). Power evacuation structure is mostly a state power utility subject and requires minimal investment.

"The project investor should choose the right engineering, procurement and construction (EPC) contractor who can skillfully design the solar power plant, supply the required components and also execute the project with good workmanship—ensuring useful life of 25 years, and undertaking operation and maintenance for at least 10 to 12 years if not 25 years," says Subramanya.

"Solar module is the key building block, both technically (as it solely contributes to the energy production of the plant) and commercially (as it constitutes over 70 per cent of the project cost). So if the EPC contractor is an established solar panel manufacturer in the country with a good financial track record and credibility that suggests dependability over next 25 years, considerable de-risking is done from the point of view of the investor in terms of securing the returns," he adds.

S.R.C. Sathyanarayan, head-PV power plants, TÜV Rheinland, lists out the financial, technical and other requirements as below:

Financial criteria. The net worth

of the company should be equal to or greater than the value calculated at the rate of Rs 30 million or equivalent US\$ per MW of the project capacity up to 20 MW. For every MW additional capacity, beyond 20 MW, additional net worth of Rs 20 million needs to be demonstrated. The net worth is computed on the basis of unconsolidated audited annual accounts of the company.

The company is required to submit annual audited accounts for the last four financial years (or starting from the year of incorporation if the period of existence of the company is less than four years), indicating the year which should be considered for evaluation, along with a net worth certificate from a chartered accountant to demonstrate fulfillment of the criteria. However, for new as well as existing companies, the net worth criteria can also be met as on date not more than seven days prior to the date of submission of request for selection (R/S) by the company.

For the purpose of meeting financial requirements, only unconsolidated audited annual accounts of the company are used. However, consolidated accounts may be used provided the project developer has at least 26 per cent equity in each company whose accounts are merged in the audited consolidated account and the financial capability of these companies is not considered again for the purpose of evaluation of the bid.

Technical criteria. Under Phase I of the JNNISM, it is proposed to promote only commercially established and operational technologies to minimise the technology risk and to achieve the commissioning of the projects. The detailed technical parameters for solar PV projects are shown in the box.

Connectivity with the grid. The plant should be designed for intercon-

Why it makes sense

1. Annual solar radiation in India is 5.2-6.5 kWh/m²/day, and thus there is an abundant stock to trap
2. Ample land available for solar parks
3. Availability of government policies both at the national level (Jawaharlar Nehru National Solar Mission) and state level (Solar Power Policy in Gujarat, Karnataka and Rajasthan)
4. Attractive tariff rates, in the range of Rs 11-18 per kWh
5. Financial assistance from financial institutions and banks
6. Reduced customs duty followed by exempted excise duty on solar PV panels
7. Carbon credit benefits
8. Payback period of six to eight years
9. Project life of 25 years

—Deepak Zade, senior vice president (energy and carbon services), MITCON Consultancy & Engineering Services